

90-Day Metastatic Cancer Diagnosis Predictor

Sophie Juco (smj490) & Elaf Almahmoud (ea3171)

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Capstone Project

Due on Thursday, May 09 at 11:59pm EST

1 Background

1.1 Purpose of the ADS & Trade-offs

The Automated Diagnostic System (ADS) aims to accurately predict whether a patient will receive a diagnosis within 90 days following testing. The ADS focuses on Metastatic Triple Negative Breast Cancer and addresses the urgent need for timely diagnosis and treatment, where delays can lead to severe outcomes due to the aggressiveness of this particular type of cancer. The ADS has no additional goals that would lead to trade-offs since it is focused on a single objective. The Google Colab Notebook containing the ADS code can be accessed here¹.

2 Input & Output

2.1 Data Details

The dataset used in this Kaggle competition was provided to Women in Data Science (WiDS) Worldwide, the host of this competition, by Gilead Sciences which is a biopharmaceutical company. The dataset from Gilead Sciences contains real-world information about patients' demographics, diagnosis and treatment options, and insurance provided for patients diagnosed with breast cancer between 2015-2018 which originates from Health Verity which is one of largest healthcare data ecosystems in the US. Gilead Sciences enriched the data by adding third party geo-demographic data to include socio-economic aspects that may contribute to health equity. For this competition, WiDS Worldwide also included ZIP code level toxicology data from NASA and Columbia University was also added onto the dataset from Gilead Sciences².

2.2 Input Features

The training data contains 82 features and 1 target variable. The test data contains the 82 features. There are 12906 rows of training data and 5792 rows of test data for a total of 18698 rows of data. The details of the training data are in the following table and the Google Colab Notebook containing the data exploration can be accessed here. The table in the Appendix contains the feature name, its description, its data type, and the number of null/missing values in the column, as well as the mean, standard deviation, minimum, 25% quartile, 50% quartile, 75% quartile and

¹Ramakrishnan, R. (2024). WiDS2024Challenge1—Baseline—V1. Kaggle.
<https://www.kaggle.com/code/ravi20076/wids2024challenge1-baseline-v1/comments?scriptVersionId=163291493>

²sk20242025, widsdatathon2024. (2024). WiDS Datathon 2024 Challenge #1. Kaggle.
<https://kaggle.com/competitions/widsdatathon2024-challenge1>

maximum (if applicable).

We look at a smaller sample of attributes below, particularly the sensitive attributes we are interested in exploring - race, age, location (region), and payer type. These attributes' data details are shown in Figure 1.

Column Name	Description	Data Type	Number of Null Values	Mean	Standard Deviation	Minimum	25% Quartile	50% Quartile	75% Quartile	Maximum
patient_race	Asian, African American, Hispanic or Latino, White, Other Race	object	6385	N/A	N/A	N/A	N/A	N/A	N/A	N/A
payer_type	payer type at Medicaid, Commercial, Medicare on the metastatic date	object	1803	N/A	N/A	N/A	N/A	N/A	N/A	N/A
patient_age	Derived from Patient Year of Birth (index year minus year of birth)	int64	0	59.18	13.34	18	50	59	67	91
region	Region of patient location	object	52	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Figure 1: Sensitive Features Data Details

For race, the entire original training data set contains 27.80% White, 8.18% Black, 6.42% Hispanic, 5.29% Other, 2.83% Asian, and 49.47% NaN. The distribution of patient race is shown in Figure 2.

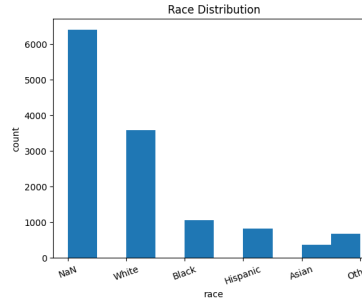


Figure 2: Patient Race Distribution

For age, when exploring fairness metrics, we split the age into senior (65+ years) and adult (<65 years). The original training dataset contains 69.49% less than 65 year olds, 30.51% 65+ year olds, and 0% NaN. The distribution of these age groups is shown in Figure 3. The overall patient age distribution before separating it into two categories is shown in Figure 4.

For region, the original training dataset contains 30.37% South, 28.94% West, 28.28% Midwest, 12.01% Northeast, and 0.40% NaN. The distribution of patient region is shown in Figure 5.

For payer type, the original training dataset contains 46.74% commercial, 19.91% medicaid, 19.39% medicare advantage, and 13.97% NaN. The distribution of payer type is shown in Figure 6.

Additionally, a correlation between features was calculated for those with relevant numerical values. The heatmap of the correlation is shown in Figure 7.

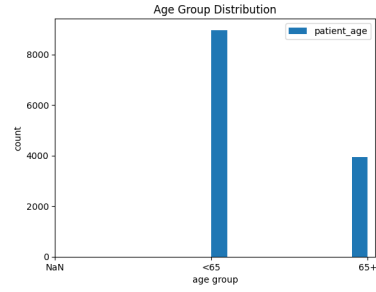


Figure 3: Patient Age (Grouped) Distribution

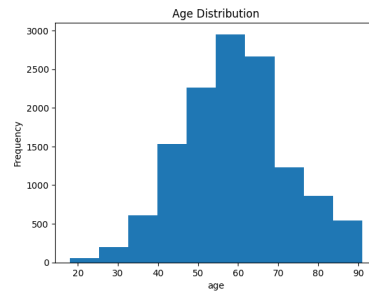


Figure 4: Patient Age Distribution

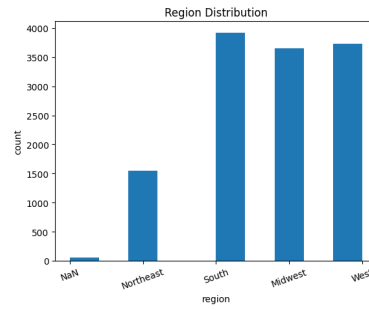


Figure 5: Patient Region Distribution

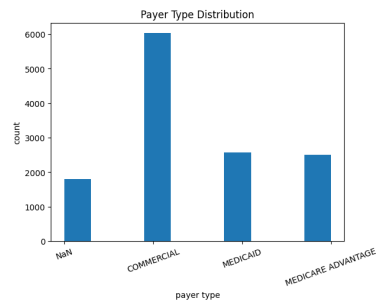


Figure 6: Patient Payer Type Distribution

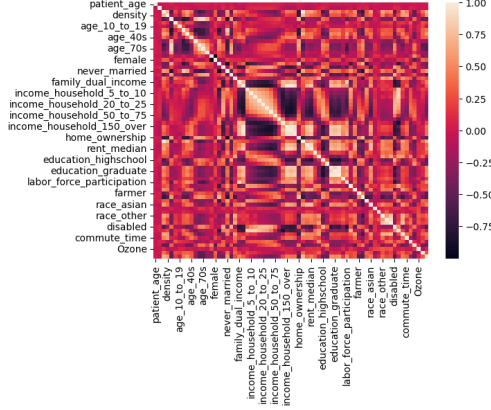


Figure 7: Correlation of Numerical Data

2.3 System Output

The output of the system is a number between 0 and 1 that indicates the likelihood of having a 1 class label, where 1 indicates that the patient will receive a diagnosis within 90 days of testing and where 0 indicates that the patient will not receive a diagnosis within 90 days of testing.

3 Implementation & Validation

3.1 Data Cleaning & Pre-processing

The code involves data pre-processing steps such as handling missing values, encoding categorical variables, and scaling features to prepare the dataset for machine learning model training. Specifically, it deals with transforming the data to ensure it is in a suitable format for applying machine learning algorithms effectively. In the Kaggle notebook for the WiDS2024 Challenge, the approach for dealing with missing values involves filling numerical columns with their median values and categorical columns with the mode.

Class preprocessor() aims to manage data pre-processing steps including reading data sets, and processing them, by first loading training and testing data from CSV files, dropping the 'patient.id' column, and standardizing column names by removing spaces and special characters. It also examines data through descriptive statistics analysis like checking unique values and nulls. A list of columns is dropped from the data, including the target variable and patient_gender, metastatic_first_novel_treatment, and metastatic_first_novel_treatment_type.

A ColumnTransformer is configured with two main transformations: (1) One-Hot Encoding: For categorical variables specified in CFG.OH_cols. It converts categorical column values into a multi-column binary representation. (2) Ordinal Encoding: For ordinal variables specified in CFG.lbl_cols. It converts categorical column values into a single column of ordinal integers. Any duplicate rows based on the feature columns are dropped. This helps in reducing redundancy and potential overfitting.

3.2 Overview of System Implementation

For the implementation of the system, the author started with simple tree based ML models and an Optuna ensemble to create sample inputs for the submission. The model training involves initializing a Random Forest Classifier with default parameters, followed by fitting it to the pre-processed training data.

The Optuna ensemble class, `OptunaEnsembler()`, defines the metric function for the competition, the objective function for the ensemble, and a method that predicts using the fitted Optuna objective.

The training pipeline elements are defined in the `MdlDeveloper()` class, where it first initializes the model predictions by initializing the CV methods and class weights. The `MdlDeveloper()` class integrates several ensemble machine learning models using Gradient Boosting methods like XGBoost (XGB1C, XGB2C, XGB3C, XGB4C), LightGBM (LGBM1C, LGBM2C, LGBM3C, LGBM4C), and CatBoost (CB1C, CB2C). Each model is tuned with parameters to optimize AUC (Area Under the Curve) metric performance.

After it trains and infers models (from the model suite), the class returns the predictions and scores. The primary metric used for model evaluation is the ROC AUC score.

3.3 Validation Methods

The ADS was validated through cross-validation based on the ROC-AUC score which we know meets the goal since it scored 0.813 which is one of the highest in the Kaggle competition.

4 Outcomes

The predictions of the ADS were numbers between 0 and 1 indicating the probability of being a 1 value since the target variable is binary - either 1 or 0. For the purposes of testing, we assumed that a predicted score of 0.5 or higher was a predicted value of 1 while a predicted score of less than 0.5 was a predicted value of 0.

The sensitive attributes we explore in particular are race, age, location, and payer type. For race, we separate the attribute into White versus non-White (NaN, Black, Hispanic, Asian, and Other) since the non-White groups are considered minority populations in the US and especially since previous studies like Obermeyer et al.'s "Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations" have shown how racial bias between Whites and Blacks arises in medical and health-care related settings, disadvantaging the Black patients³. We separated age into 65+ and less than 65 because 65+ is considered being a senior and we wanted to explore if age or senior status had an impact on diagnosis turnover time. For location, we looked at the region attribute. We separated the values into Northeast versus South, West, Midwest, and NaN under the assumption that the Northeast is typically a fairly medically progressive region with many medical facilities which could result in more awareness and more resources within the Northeast compared to the other regions. For payer type, we wanted to explore if there is a preference for what kind of insurance a patient has. We considered commercial to be the standard, regular insured healthcare

³Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage health populations. *Science* 366, 447-453.

that one would typically get through an employer while medicaid is for people with restricted income and resources and medicare is for the elderly - both of which are government funded and controlled. NaN values were also included with the medicaid and medicare group since NaN most likely indicates someone who has no health insurance at all.

4.1 Accuracy & Fairness

For our accuracy analysis across subpopulations, we will utilize the performance metrics accuracy, precision, recall, F1-score, false negative rate, and false positive rate to gauge the ADS performance on different sensitive groups. For our fairness analysis, we will employ the fairness metrics false negative rate difference, false positive rate difference, demographic parity ratio, equalized odds ratio, and selection rate difference. These metrics will help assess if the ADS treats different groups equally.

The overall accuracy of this system is 0.81. The overall precision is 0.78. The overall recall is 0.96. The overall F1 score is 0.86. The overall False Negative Rate is 0.02. The overall False Positive Rate is 0.17. The number of true negatives, false positives, false negatives, and true positives (in this order) are 803, 657, 85, and 2327. This indicates that the system as a whole is highly accurate and predicts the majority of the patients correctly. The system seems to be a little optimistic with a high false positive rate, indicating that it incorrectly predicts that a patient will be diagnosed within 90 days of testing more often than it incorrectly predicts that a patient will not be diagnosed within 90 days of testing.

For race, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.96; the overall F1-score is 0.86; the overall FNR is 0.04; and, the overall FPR is 0.45. For separating race, we separated Whites from the remaining population (NaN, Black, Hispanic, Asian, and Other). The FNR difference between Whites and the remaining population is 0.006. The FPR difference is 0.05. The demographic parity ratio is 0.94. The equalized odds ratio is 0.89. And, the selection rate difference is 0.05. The by-group accuracy scores are shown in Figure 8.

These results indicate that accuracy and fairness metrics are nearly the same between the two

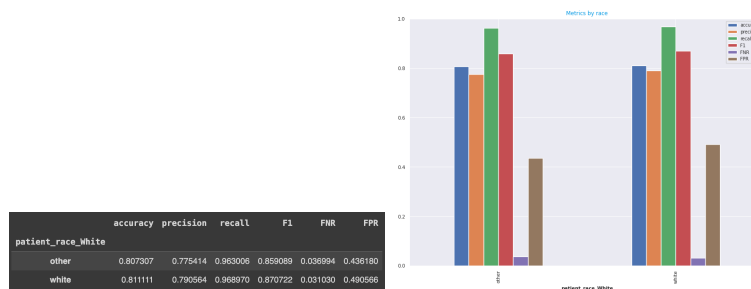


Figure 8: By-group race accuracy scores & graph

race groups. Therefore, this ADS is accurate and fair on the patients' race attribute.

Because such a large portion of race is NaN values, we wanted to see if there was a difference in results within the labeled data (aka non-NaN values). With the exclusion of NaN values, we see an overall accuracy of 0.81, an overall precision of 0.78, an overall recall of 0.97, an overall F1-score of 0.86, an overall FNR of 0.03, and an overall FPR of 0.48. For separating race, we separated Whites

from the remaining population (Black, Hispanic, Asian, and Other). The FNR difference between Whites and the remaining population is 0.003. The FPR difference is 0.02. The demographic parity ratio is 0.97. The equalized odds ratio is 0.95. And, the selection rate difference is 0.03. The by-group accuracy scores are shown in Figure 9.

These results indicate that accuracy and fairness metrics are nearly the same between the two

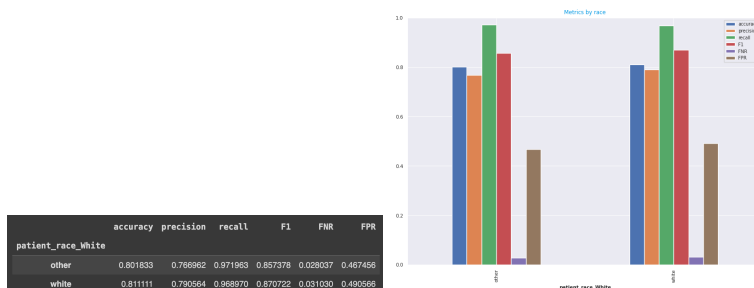


Figure 9: By-group race (no NaN values included) accuracy scores & graph

race groups, even when NaN values are removed. Therefore, this ADS is accurate and fair on the patients' race attribute.

For payer type, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.96; the overall F1-score is 0.86; the overall FNR is 0.04; and, the overall FPR is 0.45. For separating payer type, we separated COMMERCIAL from the remaining population (NaN, MEDICAID, MEDICARE ADVANTAGE) because we were interested in seeing if having no insurance (NaN) or having government assigned insurance (medicare and medicaid) would have an difference in results to “regular” healthcare plans. The FNR difference between COMMERCIAL payers and the remaining population is 0.002. The FPR difference is 0.05. The demographic parity ratio is 0.95. The equalized odds ratio is 0.90. And, the selection rate difference is 0.04. The by-group accuracy scores are shown in Figure 10.

These results indicate that accuracy and fairness metrics are nearly the same between the two

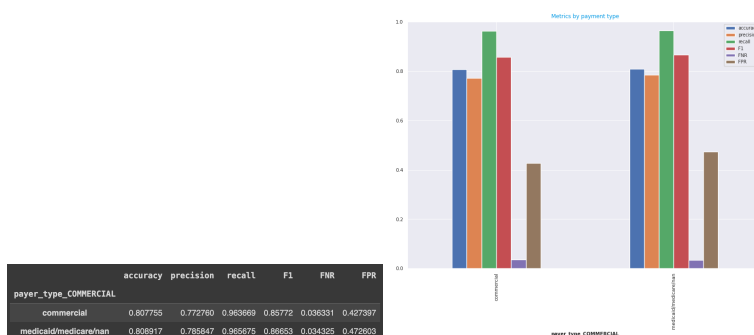


Figure 10: By-group payer type accuracy scores & graph

payer type groups. Therefore, this ADS is accurate and fair on the patients' payer type attribute.

For region, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.96; the overall F1-score is 0.86; the overall FNR is 0.04; and, the overall FPR is 0.45. For separating region, we separated Northeast from the remaining population (NaN, South, West, Midwest). The

FNR difference between Northeast and the remaining population is 0.002. The FPR difference is 0.10. The demographic parity ratio is 0.91. The equalized odds ratio is 0.78. And, the selection rate difference is 0.07. The by-group accuracy scores are shown in Figure 11.

These results indicate that the majority of accuracy and fairness metrics are nearly the same be-

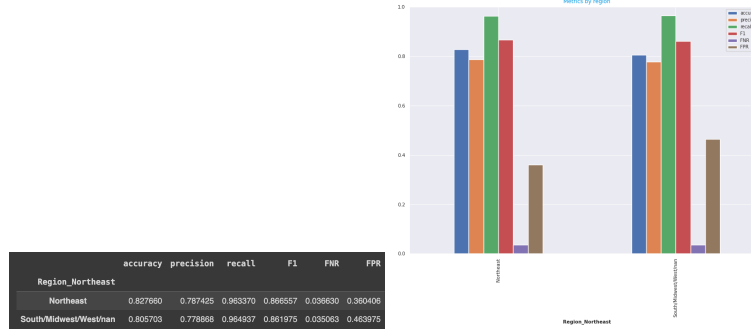


Figure 11: By-group region accuracy scores & graph

tween the two region groups. However, there is a difference in FPR between the groups with FPR being higher for the South/Midwest/West/NaN group by about 10% than the Northeast group. This also results in a lower equalized odds score than previous attributes.

For age, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.96; the overall F1-score is 0.86; the overall FNR is 0.04; and, the overall FPR is 0.45. For separating age, we separated 65+ from the remaining population (<65). The FNR difference between 65+ and the remaining population is 0.0002. The FPR difference is 0.08. The demographic parity ratio is 0.99. The equalized odds ratio is 0.82. And, the selection rate difference is 0.006. The by-group accuracy scores are shown in Figure 12.

These results indicate that accuracy and fairness metrics are nearly the same between the two age

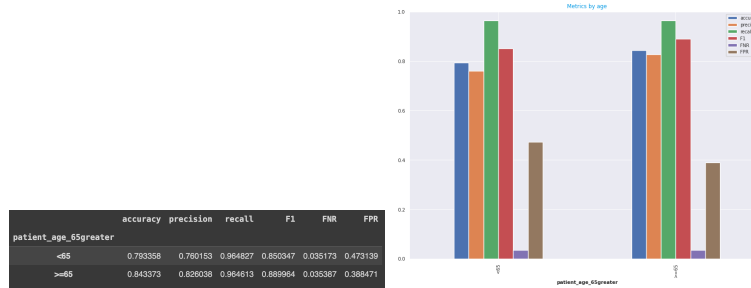


Figure 12: By-group age accuracy scores & graph

groups. Therefore, this ADS is accurate and fair on the patients' age attribute.

4.2 Robustness & Reproducibility

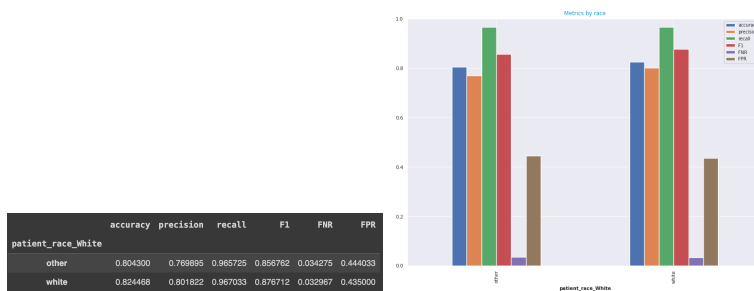
While further examining this ADS and Kaggle competition, we ran into the issue of the y_{test} data not being supplied by the competition hosts. We looked through the documentation of the competition, the host's website, and the comments/discussion on the competition and not only were we not able to find the true y_{test} values, but also the competition hosts explicitly stated that they would not release the true y_{test} values in response to a commenter asking for the data.

[illegible]

tation, discussion, and previous versions of the ADS but there was no record of where these other values came from or how to recalculate them. So, in order to test the reproducibility of this ADS on new data, we reran the ADS and the accuracy and fairness metrics on three additional data splits.

Seed=0: For seed 0, the overall accuracy is 0.81. The overall precision is 0.78. The overall recall is 0.96. The overall F1 score is 0.86. The overall False Negative Rate is 0.02. The overall False Positive Rate is 0.17. The number of true negatives, false positives, false negatives, and true positives (in this order) are 827, 654, 81, and 2310. These are nearly identical metrics as the original train/test split.

These accuracy and fairness metrics on race are extremely similar to the original train/test split,



indicating that the ADS results have been reproduced for this attribute in this dataset.

For payer type, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.97; the overall F1-score is 0.86; the overall FNR is 0.03; and, the overall FPR is 0.44. The FNR difference between COMMERCIAL payers and the remaining population is 0.008. The FPR difference is 0.03. The demographic parity ratio is 0.97. The equalized odds ratio is 0.94. And, the selection rate difference is 0.02. The by-group accuracy scores are shown in Figure 15.

These accuracy and fairness metrics on payer type are extremely similar to the original train/test

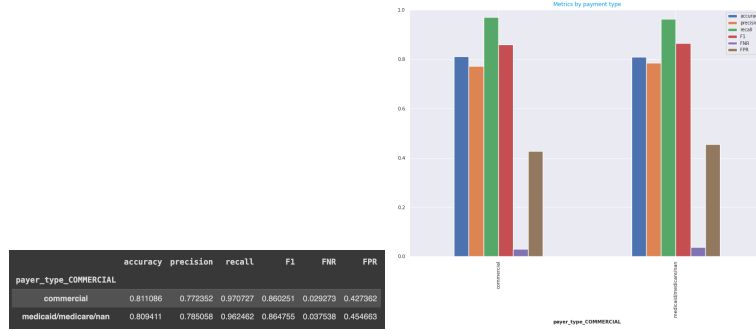


Figure 15: By-group payer type accuracy scores & graph - seed 0

split, indicating that the ADS results have been reproduced for this attribute in this dataset.

For region, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.97; the overall F1-score is 0.86; the overall FNR is 0.03; and, the overall FPR is 0.44. The FNR difference between Northeast and the remaining population is 0.003. The FPR difference is 0.03. The demographic parity ratio is 0.95. The equalized odds ratio is 0.93. And, the selection rate difference is 0.03. The by-group accuracy scores are shown in Figure 16.

These accuracy and fairness metrics on region are extremely similar to the original train/test

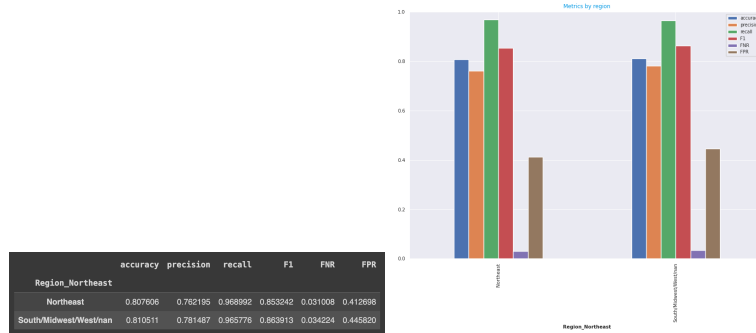


Figure 16: By-group region accuracy scores & graph - seed 0

split. The only big difference was in the equalized odds ratio which improved by about 15% in this dataset. Overall, the similarities in the metrics indicate that the ADS results have been reproduced for this attribute in this dataset.

For age, the overall accuracy is 0.81; the overall precision is 0.78; the overall recall is 0.97; the overall F1-score is 0.86; the overall FNR is 0.03; and, the overall FPR is 0.44. The FNR difference between 65+ and the remaining population is 0.008. The FPR difference is 0.09. The demographic

parity ratio is 0.97. The equalized odds ratio is 0.80. And, the selection rate difference is 0.02. The by-group accuracy scores are shown in Figure 17.

These accuracy and fairness metrics on age are extremely similar to the original train/test split,

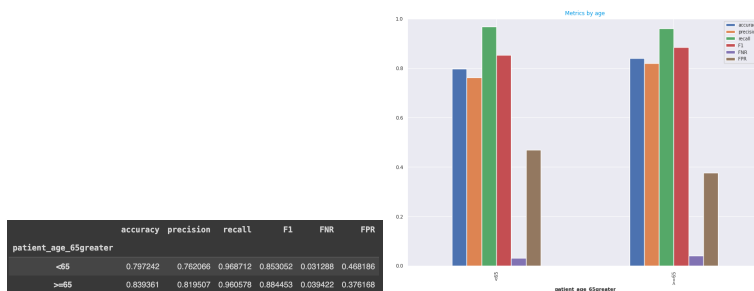


Figure 17: By-group age accuracy scores & graph - seed 0

indicating that the ADS results have been reproduced for this attribute in this dataset.

Seed=5: For seed 5, the overall accuracy is 0.82. The overall precision is 0.79. The overall recall is 0.97. The overall F1 score is 0.87. The overall False Negative Rate is 0.02. The overall False Positive Rate is 0.16. The number of true negatives, false positives, false negatives, and true positives (in this order) are 774, 634, 78, and 2386. These are nearly identical metrics as the original train/test split.

For race, the overall accuracy is 0.82; the overall precision is 0.79; the overall recall is 0.97; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between Whites and the remaining population is 0.003. The FPR difference is 0.04. The demographic parity ratio is 0.95. The equalized odds ratio is 0.91. And, the selection rate difference is 0.04. The by-group accuracy scores are shown in Figure 18.

These accuracy and fairness metrics on race are extremely similar to the original train/test split,

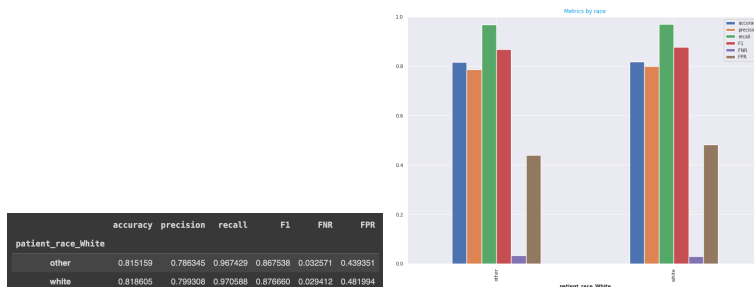


Figure 18: By-group race accuracy scores & graph - seed 5

indicating that the ADS results have been reproduced for this attribute in this dataset.

For payer type, the overall accuracy is 0.82; the overall precision is 0.79; the overall recall is 0.97; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between COMMERCIAL payers and the remaining population is 0.006. The FPR difference is 0.09. The demographic parity ratio is 0.92. The equalized odds ratio is 0.81. And, the selection rate difference is 0.06. The by-group accuracy scores are shown in Figure 19.

These majority of the accuracy and fairness metrics on payer type are extremely similar to the

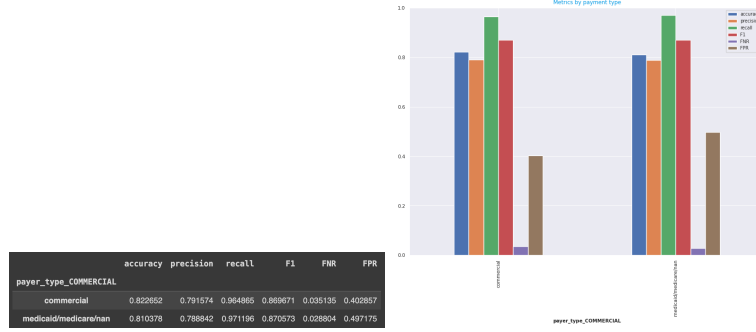


Figure 19: By-group payer type accuracy scores & graph - seed 5

original train/test split. The equalized odds ratio is slightly lower than the original value. Overall, these results indicate that the ADS results have been reproduced for this attribute in this dataset.

For region, the overall accuracy is 0.82; the overall precision is 0.79; the overall recall is 0.97; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between Northeast and the remaining population is 0.02. The FPR difference is 0.04. The demographic parity ratio is 0.99. The equalized odds ratio is 0.92. And, the selection rate difference is 0.007. The by-group accuracy scores are shown in Figure 20.

These accuracy and fairness metrics on region are extremely similar to the original train/test

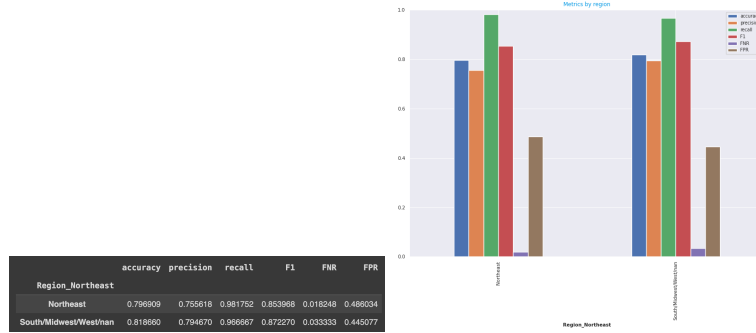


Figure 20: By-group region accuracy scores & graph - seed 5

split. The only big difference was in the equalized odds ratio which improved by about 14% in this dataset. Overall, the similarities in the metrics indicate that the ADS results have been reproduced for this attribute in this dataset.

For age, the overall accuracy is 0.82; the overall precision is 0.79; the overall recall is 0.97; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between 65+ and the remaining population is 0.0001. The FPR difference is 0.07. The demographic parity ratio is 0.998. The equalized odds ratio is 0.86. And, the selection rate difference is 0.001. The by-group accuracy scores are shown in Figure 21.

These accuracy and fairness metrics on age are extremely similar to the original train/test split, indicating that the ADS results have been reproduced for this attribute in this dataset.

Seed=20: For seed 20, the overall accuracy is 0.81. The overall precision is 0.79. The overall

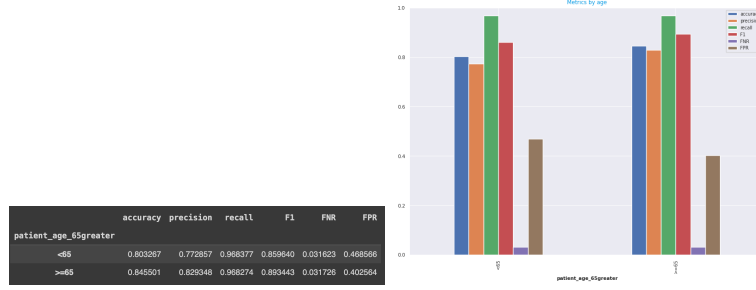


Figure 21: By-group age accuracy scores & graph - seed 5

recall is 0.96. The overall F1 score is 0.87. The overall False Negative Rate is 0.02. The overall False Positive Rate is 0.16. The number of true negatives, false positives, false negatives, and true positives (in this order) are 786, 632, 93, and 2361. These are nearly identical metrics as the original train/test split.

For race, the overall accuracy is 0.81; the overall precision is 0.79; the overall recall is 0.96; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between Whites and the remaining population is 0.006. The FPR difference is 0.04. The demographic parity ratio is 0.94. The equalized odds ratio is 0.91. And, the selection rate difference is 0.04. The by-group accuracy scores are shown in Figure 22.

These accuracy and fairness metrics on race are extremely similar to the original train/test split,

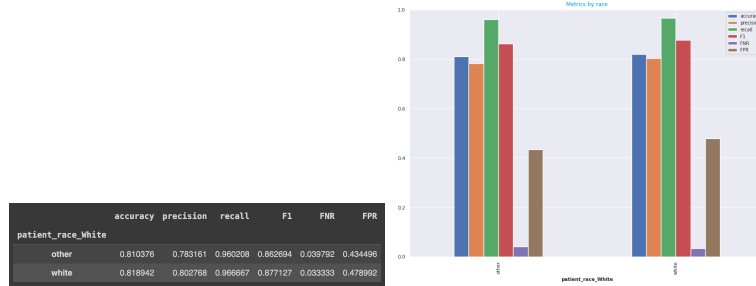


Figure 22: By-group race accuracy scores & graph - seed 20

indicating that the ADS results have been reproduced for this attribute in this dataset.

For payer type, the overall accuracy is 0.81; the overall precision is 0.79; the overall recall is 0.96; the overall F1-score is 0.87; the overall FNR is 0.03; and, the overall FPR is 0.45. The FNR difference between COMMERCIAL payers and the remaining population is 0.01. The FPR difference is 0.04. The demographic parity ratio is 0.95. The equalized odds ratio is 0.82. And, the selection rate difference is 0.04. The by-group accuracy scores are shown in Figure 23.

These accuracy and fairness metrics on payer type are extremely similar to the original, indicating that the ADS results have been reproduced for this attribute in this dataset.

For region, the overall accuracy is 0.81; the overall precision is 0.79; the overall recall is 0.96; the overall F1-score is 0.87; the overall FNR is 0.04; and, the overall FPR is 0.45. The FNR difference between Northeast and the remaining population is 0.003. The FPR difference is 0.02. The demographic parity ratio is 0.94. The equalized odds ratio is 0.96. And, the selection rate

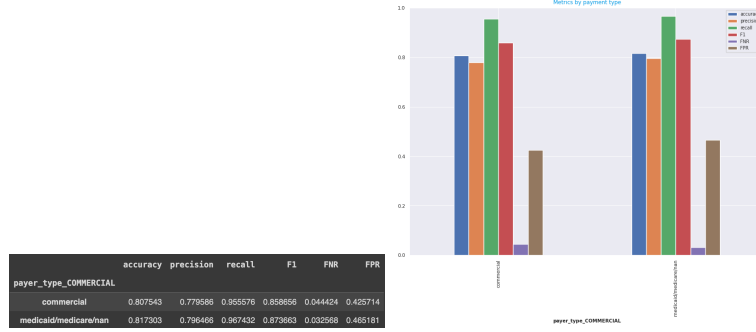


Figure 23: By-group payer type accuracy scores & graph - seed 20

difference is 0.05. The by-group accuracy scores are shown in Figure 24.

These accuracy and fairness metrics on region are extremely similar to the original train/test

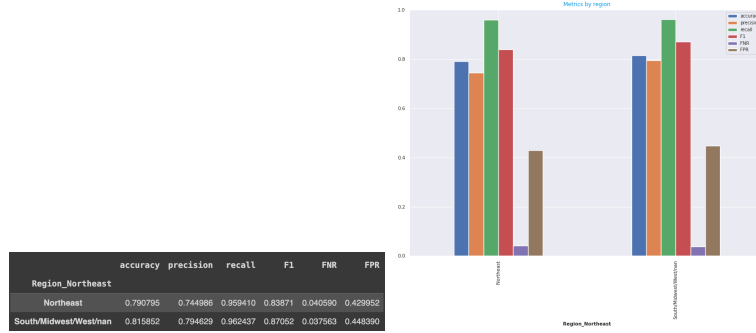


Figure 24: By-group region accuracy scores & graph - seed 20

split. The only big difference was in the equalized odds ratio which improved by about 18% in this dataset. Overall, the similarities in the metrics indicate that the ADS results have been reproduced for this attribute in this dataset.

For age, the overall accuracy is 0.81; the overall precision is 0.79; the overall recall is 0.96; the overall F1-score is 0.87; the overall FNR is 0.04; and, the overall FPR is 0.45. The FNR difference between 65+ and the remaining population is 0.002. The FPR difference is 0.07. The demographic parity ratio is 0.99. The equalized odds ratio is 0.85. And, the selection rate difference is 0.01. The by-group accuracy scores are shown in Figure 25.

These accuracy and fairness metrics on age are extremely similar to the original train/test split, indicating that the ADS results have been reproduced for this attribute in this dataset.

5 Summary

5.1 Data Conclusions

Although the data is not perfectly balanced, it generally provides a reasonable and appropriate framework for this ADS. Since the dataset includes sensitive attributes like race, age, payer type, and region, it is crucial to analyze these variables to achieve our goals, such as assessing disparities across subgroups. However, the presence of NaN values, particularly in race and insurance type, may introduce biases or uncertainty if the missing values are not random

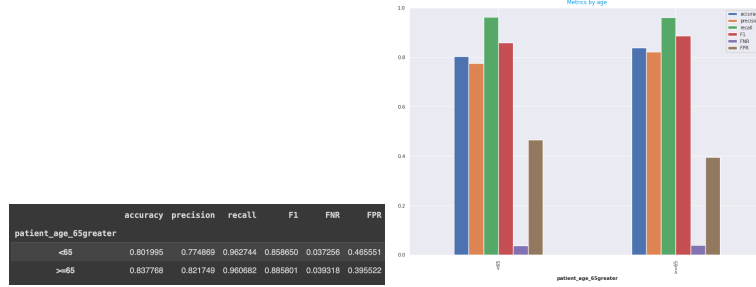


Figure 25: By-group age accuracy scores & graph - seed 20

5.2 Implementation Conclusions

Given the following accuracy metric scores: an overall accuracy of approximately 0.81 consistently across all seeds, a precision of around 0.78, a recall of about 0.96, and an F1 score of approximately 0.86, alongside the fairness metrics that show a demographic parity ratio close to 1.0 across attributes, an equalized odds ratio ranging from around 0.78 to 0.98, and false positive/negative rate differences that are typically low (less than 0.1), the implementation appears robust, accurate, and reasonably fair based on the metrics used and the detailed evaluation across multiple seeds and sensitive attributes.

Stakeholders who may find these measures appropriate include healthcare providers and policy-makers interested in understanding disparities among different patient demographics and insurance types, as well as patients and society at large who would find consistent treatment across racial, age, and economic lines important.

5.3 ADS in the Public Sector

Although the model demonstrates consistent accuracy and fairness across seeds and subgroups, which is crucial for making reliable predictions and reducing biases, certain groups (e.g., age and region) still exhibit noticeable false positive/negative rate differences. Given the high-stakes nature of this model's decisions, it necessitates an exceptionally reliable and accurate model. Additionally, as we found, there are issues in the transparency of the predictions of the original ADS in that they incorporate some previous calculation that is not documented or reproducible. Therefore, we would not be comfortable deploying this ADS as is.

5.4 Improvements

One suggestion to improve the data is to increase representation by including more underrepresented groups and to incorporate more effective data processing techniques, such as handling missing values more effectively using fairer imputation methods. Regarding methodology, using cross-validation could enhance robustness rather than relying on a fixed train-test split. Furthermore, the ADS should provide explanations through interpretability tools like SHAP and LIME to clarify how predictions are being generated and what features they are focusing on.

References

- [1] Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage health populations. *Science* 366, 447-453.

[2] Ramakrishnan, R. (2024). WiDS2024Challenge1—Baseline—V1. Kaggle. <https://www.kaggle.com/code/ravi20076/wids2024challenge1-baseline-v1/comments?scriptId=163291493>

[3] sk20242025, widsdatathon2024. (2024). WiDS Datathon 2024 Challenge #1. Kaggle. <https://kaggle.com/competitions/widsdatathon2024/challenge1>

A Python Notebooks

The code hosted in [THIS](#) Google Colab Notebook contains the analysis of the data given in the Kaggle competition.

The code hosted in [THIS](#) Google Colab Notebook contains 3 sections. The first section is the author's original code, slightly modified to run through Google Colab. The second part contains our initial accuracy and fairness analysis. The third part contains the same accuracy and fairness analyses on three additional data splits.

B Additional Data Details

The below table shows the complete collection of import features and their details:

Column Name	Description	Data Type	Number of Null Values	Mean	Standard Deviation	Minimum	25% Quartile	50% Quartile	75% Quartile	Maximum
patient_id	Unique identification number of patient	int64	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
patient_race	Asian, African American, Hispanic or Latino, White, Other Race	object	6385	N/A	N/A	N/A	N/A	N/A	N/A	N/A
payer_type	payer type at Medicaid, Commercial, Medicare on the metastatic date	object	1803	N/A	N/A	N/A	N/A	N/A	N/A	N/A
patient_state	Patient State (e.g. AL, AK, AZ, AR, CA, CO etc...) on the metastatic date	object	51	N/A	N/A	N/A	N/A	N/A	N/A	N/A
patient_zip3	Patient Zip3 (e.g. 190) on the metastatic date	int64	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
patient_age	Derived from Patient Year of Birth (index year minus year of birth)	int64	0	59.18	13.34	18	50	59	67	91
patient_gender	F, M on the metastatic date	object	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
bmi	If Available, will show available BMI information (Earliest BMI recording post metastatic date)	float64	8965	28.98	5.7	14	24.66	28.19	32.92	85
breast_cancer_diagnosis_code	ICD10 or ICD9 diagnoses code	object	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
breast_cancer_diagnosis_desc	ICD10 or ICD9 code description. This column is raw text and may require NLP/processing and cleaning	object	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
metastatic_cancer_diagnosis_code	ICD10 diagnoses code	object	0	N/A	N/A	N/A	N/A	N/A	N/A	N/A

metastatic_first_novel_treatment	Generic drug name of the first novel treatment (e.g. "Cisplatin") after metastatic diagnosis	object	12882	N/A	N/A	N/A	N/A	N/A	N/A	N/A
metastatic_first_novel_treatment_type	Description of Treatment (e.g. Antineoplastic) of first novel treatment after metastatic diagnosis	object	12882	N/A	N/A	N/A	N/A	N/A	N/A	N/A
region	Region of patient location	object	52	N/A	N/A	N/A	N/A	N/A	N/A	N/A
division	Division of patient location	object	52	N/A	N/A	N/A	N/A	N/A	N/A	N/A
population	An estimate of the zip code's population.	float64	1	20744.44	13886.9	635.55	9463.9	19154.19	30021.28	71374.13
density	The estimated population per square kilometer.	float64	1	1581.95	2966.31	0.92	171.86	700.34	1666.52	21172
age_median	The median age of residents in the zip code.	float64	1	40.5	4.04	20.6	37.13	40.64	42.93	54.57
age_under_10	The percentage of residents aged 0-9.	float64	1	11.12	1.51	0	10.16	11.04	12.19	17.68
age_10_to_19	The percentage of residents aged 10-19.	float64	1	12.95	1.92	6.31	11.74	12.92	14.02	35.3
age_20s	The percentage of residents aged 20-29.	float64	1	13.29	3.35	5.93	11.01	12.54	14.97	62.1
age_30s	The percentage of residents aged 30-39.	float64	1	12.86	2.32	1.5	11.29	12.44	14.03	25.47
age_40s	The percentage of residents aged 40-49.	float64	1	12.07	1.25	0.8	11.34	12.12	12.89	17.82
age_50s	The percentage of residents aged 50-59.	float64	1	13.44	1.64	0	12.3	13.57	14.58	21.66
age_60s	The percentage of residents aged 60-69.	float64	1	12.62	2.57	0.2	10.62	12.53	13.99	29.85
age_70s	The percentage of residents aged 70-79.	float64	1	7.65	2.15	0	6.01	7.32	8.79	19

age_over_80	The percentage of residents aged over 80.	float64	1	4	1.24	0	3.28	3.8	4.53	18.83
male	The percentage of residents who report being male (e.g. 55.1).	float64	1	50.1	1.66	39.73	49.16	49.98	50.94	61.6
female	The percentage of residents who report being female (e.g. 44.9).	float64	1	49.9	1.66	38.4	49.06	50.02	50.84	60.28
married	The percentage of residents who report being married (e.g. 44.9).	float64	1	47.68	7.49	0.9	42.93	49.43	52.99	66.9
divorced	The percentage of residents divorced.	float64	1	12.67	2.05	0.2	11.16	12.65	14.19	19.83
never_married	The percentage of residents never married.	float64	1	33.83	8.02	13.44	27.53	32	39.23	98.9
widowed	The percentage of residents never widowed.	float64	1	5.81	1.54	0	4.76	5.52	6.52	23.05
family_size	The average size of resident families (e.g. 3.22).	float64	4	3.2	0.22	2.55	3.04	3.17	3.35	4.17
family_dual_income	The percentage of families with dual income earners.	float64	4	51.85	6.81	19.31	47.64	52.59	56.61	70.93
income_household_median	Median household income in USD.	float64	4	74374.37	20712.59	29222	61284.45	69803.13	83539	164119.18
income_household_under_5	The percentage of households with income under \$5,000.	float64	4	3.27	1.45	0.75	2.26	2.84	4.01	19.62
income_household_5_to_10	The percentage of households with income from \$5,000-\$10,000.	float64	4	2.51	1.34	0.36	1.54	2.16	3.11	11.87
income_household_10_to_15	The percentage of households with income from \$10,000-\$15,000.	float64	4	4.14	1.77	1.02	2.9	3.72	5.08	14.28
income_household_15_to_20	The percentage of households with income from \$15,000-\$20,000.	float64	4	3.93	1.45	1.03	2.92	3.77	4.62	12.92

income_household_20_to_25	The percentage of households with income from \$20,000-\$25,000.	float64	4	4.07	1.35	1.1	3.17	4.04	4.81	14.35
income_household_25_to_35	The percentage of households with income from \$25,000-\$35,000.	float64	4	8.4	2.21	2.65	6.78	8.44	9.91	18.34
income_household_35_to_50	The percentage of households with income from \$35,000-\$50,000.	float64	4	11.56	2.58	1.7	9.9	11.79	13.33	24.08
income_household_50_to_75	The percentage of households with income from \$50,000-\$75,000.	float64	4	16.89	2.72	4.95	15.34	17.08	18.53	27.13
income_household_75_to_100	The percentage of households with income from \$75,000-\$100,000.	float64	4	12.66	1.83	4.73	11.35	12.68	13.95	24.8
income_household_100_to_150	The percentage of households with income from \$100,000-\$150,000.	float64	4	15.83	3.18	4.29	13.62	16.02	18.42	31.33
income_household_150_over	The percentage of households with income over \$150,000.	float64	4	16.72	8.94	0.84	10.07	14.7	21.32	52.82
income_household_six_figure	Percentage of households that earn at least \$100,000 (e.g. 25.3)	float64	4	32.55	11.13	5.69	24.58	30.58	39.56	69.03
income_individual_median	The median income of individuals in the zip code.	float64	1	36520.52	8195.16	4316	31575.26	35253.09	39948.52	88910.5
home_ownership	Percentage of households that own (rather than rent) their residence.	float64	4	65.99	14.12	15.85	56.53	69.67	77.19	90.37
housing_units	The number of housing units (or households) in the zip code.	float64	1	7575.71	4916.91	0	3446.64	6994.41	10939.08	25922.55
home_value	The median value of homes that are owned by residents.	float64	4	3.40E+05	2.52E+05	6.06E+04	1.68E+05	2.48E+05	4.06E+05	1.85E+06

rent_median	The median rent paid by renters.	float64	4	1237.3	427.84	448.4	895.79	1168	1506.4	2965.25
rent_burden	The median rent as a percentage of the median renter's household income.	float64	4	31.34	4.78	17.42	28.2	30.99	33.81	78.94
education_less_highschool	The percentage of residents with less than a high school education.	float64	1	11.99	5.14	0	8.25	10.84	15.34	34.33
education_highschool	The percentage of residents with a high school diploma but no more.	float64	1	27.56	7.99	0	21.64	27.41	33.41	53.96
education_some_college	The percentage of residents with some college but no more.	float64	1	28.92	4.89	7.2	26.03	29.29	32.37	50.13
education_bachelors	The percentage of residents with a bachelor's degree (or equivalent) but no more.	float64	1	19.27	6.17	2.47	14.02	19.05	23.89	41.7
education_graduate	The percentage of residents with a graduate degree.	float64	1	12.26	5.94	2.09	7.65	10.8	15.86	51.84
education_college_or_above	The percentage of residents with at least a 4-year degree.	float64	1	31.53	11.63	7.05	22.09	30.14	39.18	77.82
education_stem_degree	The percentage of college graduates with a Bachelor's degree or higher in a Science and Engineering (or related) field.	float64	1	43.39	4.64	23.91	40.07	43.07	46.21	73
labor_force_participation	The percentage of residents 16 and older in the labor force.	float64	1	61.62	5.95	30.7	57.94	62.78	65.73	78.67
unemployment_rate	The percentage of residents unemployed.	float64	1	5.95	1.95	0.82	4.72	5.47	6.84	18.8
self_employed	The percentage of households reporting self-employment	float64	4	13.21	3.41	2.26	10.87	12.75	15.22	25.54

	income on their 2016 IRS tax return.									
farmer	The percentage of households reporting farm income on their 2016 IRS tax return.	float64	4	1.86	3.06	0	0.04	0.45	2.32	26.73
race_white	The percentage of residents who report their race White.	float64	1	69.72	17.96	14.5	56.76	70.88	85.58	98.44
race_black	The percentage of residents who report their race as Black or African American.	float64	1	11.45	12.53	0.06	2.24	6.41	16.22	69.66
race_asian	The percentage of residents who report their race as Asian.	float64	1	5.47	6.74	0	1.1	2.97	7.07	49.85
race_native	The percentage of residents who report their race as American Indian and Alaska Native.	float64	1	0.9	2.5	0	0.21	0.43	0.79	76.93
race_pacific	The percentage of residents who report their race as Native Hawaiian and Other Pacific Islander.	float64	1	0.14	0.51	0	0.02	0.05	0.15	14.76
race_other	The percentage of residents who report their race as Some other race.	float64	1	5.68	6.25	0.003	1.31	3.51	7.92	33.19
race_multiple	The percentage of residents who report their race as Two or more races.	float64	1	6.64	3.54	0.43	3.92	5.8	8.85	26.43
hispanic	The percentage of residents who report being Hispanic. Note: Hispanic is considered to be an ethnicity and not a race.	float64	1	18.46	17.03	0.19	4.71	11.98	27.99	91

disabled	The percentage of residents who report a disability.	float64	1	13.34	3.69	4.6	10.27	12.88	15.56	35.16
poverty	The median value of owner occupied homes.	float64	4	13.41	5.22	3.43	9.66	12.18	16.64	38.35
limited_english	The percentage of residents who only speak limited English.	float64	4	4.47	4.84	0	0.99	2.75	5.98	26.76
commute_time	The median commute time of resident workers in minutes.	float64	1	27.98	5.08	12.46	24.93	27.79	30.71	48.02
health_uninsured	The percentage of residents who report not having health insurance.	float64	1	8.58	4.2	2.44	5.62	7.47	10.62	27.57
veteran	The percentage of residents who are veterans.	float64	1	7.08	3.11	1.2	4.93	6.85	8.62	25.2
ozone	Annual Ozone (O3) concentration data at Zip3 level. This data shows how air quality data may impact health.	float64	29	39.82	3.56	30.94	37.7	39.11	41.14	42.24
PM25	Annual Fine Particulate Matter (PM2.5) concentration data at Zip3 level. This data shows how air quality data may impact health.	float64	29	7.48	1.52	2.64	6.65	7.69	8.28	11.17
N02	Annual Nitrogen Dioxide (NO2) concentration data at Zip3 level. This data shows how air quality data may impact health.	float64	29	16.1	5.84	2.76	11.28	15.59	20.8	31.5